

LAB 8 REPORT

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Lab Title: Data Type, If -Elif Statement, Operators

Theory: Data Types: Data types define the type of data stored in a variable, such as numbers, strings, lists, and booleans.

If-Elif Statement: It is used for decision-making in Python by checking conditions and executing code based on the result.

Operators: Operators are symbols or keywords used to perform operations like arithmetic, comparison, and logical calculations.

TASK 1

```
# DATA TYPES
```

```
# EXAMPLE 1
```

```
NUM=int(input("Enter a number: "))  
print(type(NUM))
```

```
# EXAPMLE 2
```

```
name=input("Enter your name: ")  
print(type(name))
```

Output:

```
Enter a number: 25  
<class 'int'>  
Enter your name: sohaib omix  
<class 'str'>  
  
Process finished with exit code 0
```

TASK 2

```
# Task 2
```

```
# EXAMPLE 1
```

```
obtained_marks=int(input("enter the number of obtained marks:"))
```

```

if obtained_marks >= 90:
    print("Grade A+:")
elif obtained_marks >= 85:
    print("Grade A:")
elif obtained_marks >= 80:
    print("Grade A-:")
elif obtained_marks >= 75:
    print("Grade B+:")
elif obtained_marks >= 70:
    print("Grade B:")
elif obtained_marks >= 65:
    print("Grade B-:")
elif obtained_marks >= 60:
    print("Grade C+:")
elif obtained_marks >= 55:
    print("Grade C:")
elif obtained_marks >= 50:
    print("Grade C-:")
elif obtained_marks << 50:
    print("Grade F:")

```

Output:

```

↓ enter the number of obtained marks: 70
Grade B:

```

EXAPMLE 2

```

obtained_marks = int(input("Enter obtained marks:"))
total_marks = 300
if obtained_marks < 0 or obtained_marks > 300:
    print("invalid choice:")
else:
    percentage = (obtained_marks/total_marks)*100
    print(percentage)
    if percentage >= 90:
        print("Grade A+:")
    elif percentage >= 85:
        print("Grade A-:")
    elif percentage >= 80:
        print("Grade B+:")

```

```

elif percentage >= 75:
    print("Grade B-:")
elif percentage >= 70:
    print("Grade C+:")
elif percentage >= 65:
    print("Grade C-:")
elif percentage >= 60:
    print("Grade D+:")
elif percentage <60:
    print("Grade F:")

```

Output:

```

Enter obtained marks:200
66.66666666666666
Grade C-:

Process finished with exit code 0

```

TASK 3

```
# OPERATORS
```

```
# EXAMPLE 1
```

```

A=10
B=22
print(A+B)
print(A-B)

```

Output:

```

C:\Users\Suleima
32
-12

```

```
# EXAMPLE 2
```

```

a=56
b=4
c=a%b
print(c)

```

```
d=a+b  
print(d)  
e=a/b  
print(e)  
f=a//b  
print(f)
```

Output:

```
0  
60  
14.0  
14  
  
Process finished with exit code 0
```